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COSIMA Cookbook: a community-driven resource for ocean and sea-ice modelling science

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Software

- Review
- Repository
- Archive

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Summary

The COSIMA Cookbook is an open educational resource for analysing ocean and sea-ice model output in Jupyter notebooks (Kluyver et al., 2016). It has been developed by the Consortium for Ocean–Sea Ice Modelling in Australia (COSIMA; https://cosima.org.au) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean–sea ice model configurations (e.g. Kiss et al., 2020). The Cookbook is developed openly in a GitHub repository, with all content exposed through a documentation site built with Sphinx (Komiya et al., 2025); see Figure 1. Tagged versions of the repository are uploaded to a citable archived release (Constantinou et al., 2026).

The Cookbook grew out of a loosely organised collection of Python scripts and Jupyter notebooks — many of which were broken. During the first COVID-19 lockdown in 2020, the community held its first hackathon (Martínez-Moreno & Constantinou, 2020): a full-day online sprint to bring those notebooks up to date. Annual workshops and hackathons since then have gradually shaped the repository into its current curated state.

Figure 1: The COSIMA Cookbook documentation website, showing the browsable Sphinx-based landing page used to navigate tutorials and recipes. The website lives at https://cosima-recipes.readthedocs.io.

The COSIMA Cookbook is deliberately organized using a gastronomy theme to help users quickly understand and navigate the website. "Cooking Tutorials" refers to tutorials that teach generic, transferable techniques for handling ocean–sea ice model output. These tutorials give the users the foundations for understanding the analysis done in the examples. Then comes the main part of any cookbook – its recipes. Here, by "recipes" we mean self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. First come the "Easy Recipes" that provide a good starting point after the tutorials; then the "Advanced Recipes" that are more elaborate and demonstrate advanced analysis techniques. Lastly, the "Regional Specialties" contains recipes for regional model configurations (Barnes et al., 2024). This gastronomy framing is shared by other computational workflow collections, such as Project Pythia Cookbooks for the general geoscience community and the FAIR Cookbook for working with life science data.

Pedagogy is central to the project. "Cooking Tutorials" introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused recipes giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions. The recipes provide the starting point for more elaborate analyses. The categories of "easy" and "advanced" recipes provide a lightweight pedagogical cue about expected complexity and scope

Within this structure, the tutorials leverage and demonstrate open source tools for scientific analysis of large data. This includes examples of loading model output through Intake-based catalogues (Intake developers, 2026), using Xarray (Hoyer & Hamman, 2017) and Dask (Dask Development Team, 2016; Rocklin, 2015) to process very large multi-dimensional datasets in parallel. The recipe collection includes a Lagrangian particle-tracking workflow built with Parcels (Delandmeter & Sebille, 2019), as well as analyses that use xgcm (Abernathey et al., 2022) to handle variables defined on staggered finite-volume grids, and xESMF (Zhuang et al., 2025) for regridding model output. Figures are generated using matplotlib (Hunter, 2007) incorporating Cartopy (Elson et al., 2024) for map-based visualisation.

The Cookbook grew from a practical need inside the COSIMA community (COSIMA Community, 2026): powerful tools for analysing modern ocean model output already existed, but the gap between package-level documentation and reproducible end-to-end workflows left new users largely on their own. Productive use requires fluency not only in Python and Jupyter (Kluyver et al., 2016), but also in navigating high-dimensional, often very large model output, shared high-performance computing environments, and domain-specific analysis conventions. The Cookbook bridges this gap by packaging reusable workflows in the same medium in which researchers actually work.

Statement of Need

Analysing or configuring climate models carries a steep entry cost. New users must learn how to find datasets, load them efficiently, interpret metadata, operate on multi-dimensional arrays, and produce scientifically meaningful diagnostics. General-purpose libraries such as Xarray (Hoyer & Hamman, 2017) provide the foundations for these tasks, but they do not by themselves show a learner how to translate a research question into a robust analysis workflow for a specific model and computing environment. The Cookbook serves as a learning tool for those new to ocean and sea ice science, showing how to interrogate model output and visualise commonly used diagnostics — skills that transfer broadly across models and computing environments.

The gap is not only pedagogical; it directly slows science. Before any analysis can begin, a newcomer typically spends substantial time deciphering the structure of the output, the relevant conventions, and the computational environment. There is also substantial scientific labour embedded in many common diagnostics: some calculations require non-trivial development, understanding of model numerics, and validation before they can be used with confidence — these are the "Advanced Recipes". If every new user had to reconstruct those workflows independently, enormous effort would be spent rebuilding analysis code rather than doing science.

The COSIMA Cookbook facilitates knowledge sharing and accelerates research with a domain-specific, openly maintained collection of computational lessons and examples. Its contribution is not a new analysis library; rather, it is a reusable scientific and educational resource that exposes complete, well-documented workflows in the same notebook format in which many researchers actually explore data and communicate methods. By choosing notebooks instead of packaging these examples only as a software library, the project makes intermediate reasoning, methodological choices, and practical execution details visible to learners while still giving more experienced users analysis patterns they can adapt directly for research. This aligns well with JOSE's emphasis on open educational resources that enable computational learning through authentic practice.

Several design decisions have made the Cookbook particularly useful for adoption by other groups. First, each notebook is intended to be self-contained and well-documented, so learners can read, run, and modify a complete workflow without reconstructing missing context from scattered notes. Second, the repository distinguishes tutorials from recipes. Tutorials teach transferable skills, while recipes demonstrate how those skills combine in realistic analysis tasks. Third, the project includes contributor guidance and notebook review conventions that encourage new analyses to be written in a reusable and pedagogically clear style. A caveat is that most recipes use high-resolution model output with datasets of order terabytes, so the underlying data are not always practical to distribute or mirror in full. However, beyond the data-loading step, which usually early on in each recipe, the analysis workflows are generally applicable to most operational ocean–sea ice model outputs that can be loaded with Xarray (Hoyer & Hamman, 2017).

Thus, the design renders the Cookbook valuable beyond the immediate COSIMA context. Many research communities maintain model output on shared infrastructure and face the same challenge: turning expert tacit knowledge into examples that newcomers can adapt. The Cookbook offers a model for how a scientific collaboration can capture that knowledge in version-controlled notebooks, publish it as a living open resource online, and continuously improve it through community contribution.

The rise of AI-assisted coding tools adds a further dimension to the Cookbook's value — and arguably makes it more important, not less. Such tools can rapidly generate analysis code, and they produce far better results when given a well-structured, domain-specific recipe as context rather than starting from scratch. But the deeper point is this: as AI-generated code becomes routine, the critical skill shifts from *writing* code to *understanding and evaluating* it. A researcher who has worked through the Cookbook knows why a workflow is structured the way it is — which parameters matter, where scientific assumptions enter, and what a plausible result looks like. That understanding is what allows AI-generated code to be trusted, corrected, and adapted for specific science questions. Resources that teach the reasoning behind an analysis, not just its mechanics, therefore grow more valuable alongside AI adoption.

Educational Design and Experience of Use

As we have elaborated above, the learning experience is organised as a progression. The tutorials and recipes showcase Jupyter-based analysis that can be run directly on Australia's National Computational Infrastructure, and include guidance on running notebooks through JupyterLab sessions and on accessing shared data holdings. This makes the Cookbook more than a static collection of examples: it is operational documentation for a real analysis environment. At the same time, the notebooks demonstrate broadly transferable workflows for working with labelled geophysical data, so individual lessons can be reused or adapted for specific research questions or for use on other computational platforms.

The project also supports community authorship. Contributors are encouraged to submit new notebooks through pull requests, document their methods clearly, and generalise workflows so that they remain useful to others. This is particularly valuable as it is relatively rare for the computer code underlying scientific analyses to be peer reviewed with the same care as the manuscript itself. The COSIMA community provides several examples of the recipe ecosystem being extended into full research projects built on peer-authored software repositories on Github, where analysis workflows, methods, and implementation choices are exposed to community scrutiny and reuse. In that sense, the Cookbook functions both as a learning resource for end users and as a framework for teaching reproducible scientific communication through notebook design and peer review. During regularly organised hackathons, the commmunity reviews, updates, and extends existing recipes, ensuring the Cookbook remains a living resource that keeps pace with evolving tools and community best practices. The easy access to data and analysis provided by the COSIMA Cookbook has enabled rapid adoption of ACCESS ocean and sea-ice model configurations internationally — not only within physical oceanography but also across adjacent disciplines such as marine ecology(Fierro-Arcos et al., 2023), where researchers have used it as an entry point to ACCESS model output. This accessibility has facilitated well over 100 peer-reviewed papers and more than 20 PhD projects to completion to date.

Author Order

The first two authors contributed equally; authors from the third position onward are listed alphabetically by surname.

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